Chapter 24 — The Mathematical Meaning of the C & G Harmonic

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24.1 Purpose and scope

The C & G Harmonic is defined as a coupled-field framework in which $C(x,t)$ represents the measurable concentration of dispersed matter and $G(x,t)$ represents a proposed geometric-harmonic field. The concentration sector is built from established continuum transport mathematics. The geometric-harmonic sector and its coupling to C are explicitly presented as hypotheses requiring independent physical definition, dimensional calibration, and experimental testing.

The central distinction is

$$C(x,t) = \text{concentration of dispersed matter}$$

and

$$G(x,t) = \text{proposed geometric-harmonic organising field}.$$

The combined state is written abstractly as

$$H_{CG} = B_{CG}[C,G],$$

where B_{CG} is a coupling operator whose precise form must be selected according to the physical meaning and units assigned to G.

24.2 Concentration field C

24.2.1 Local conservation law

Let $C(x,t)$ denote concentration, J_C the associated flux, and S_C a source or sink. Conservation of matter gives

$$\frac{\partial C}{\partial t} + \nabla \cdot J_C = S_C.$$

For a fixed control volume V with boundary ∂V and outward normal n,

$$\frac{d}{dt}\int_V C\,dV + \oint_{\partial V} J_C \cdot n\,dA = \int_V S_C\,dV.$$

Equation (24.2) is the differential form of Equation (24.3).

24.2.2 Fickian constitutive law

For isotropic diffusion with scalar diffusivity D,

$$J_C = -D\nabla C.$$

The negative sign indicates transport from regions of high concentration toward regions of low concentration. Substitution into the conservation law gives

$$\frac{\partial C}{\partial t} = \nabla\cdot(D\nabla C) + S_C.$$

If D is constant,

$$\frac{\partial C}{\partial t} = D\nabla^2 C + S_C.$$

24.2.3 Spherical diffusion

For spherical symmetry, $C = C(r,t)$ and the angular derivatives vanish. The diffusion equation becomes

$$\frac{\partial C}{\partial t} = \frac{1}{r^2}\frac{\partial}{\partial r}\left(Dr^2\frac{\partial C}{\partial r}\right) + S_C(r,t).$$

For constant D and no source outside the emitting region,

$$\frac{\partial C}{\partial t} = D\left(\frac{\partial^2 C}{\partial r^2} + \frac{2}{r}\frac{\partial C}{\partial r}\right).$$

The radial flux is

$$J_{C,r} = -D\frac{\partial C}{\partial r}.$$

24.2.4 Steady spherical source

At steady state, $\partial C/\partial t = 0$. For constant D and $S_C = 0$ outside the source,

$$\frac{1}{r^2}\frac{d}{dr}\left(r^2\frac{dC}{dr}\right) = 0.$$

Integrating twice gives

$$C(r) = A + \frac{B}{r}.$$

If $C \to C_\infty$ as $r \to \infty$ and the source emits mass at rate $\dot{M}$,

$$C(r)=C_\infty+\frac{\dot{M}}{4\pi D r}.$$

The corresponding outward radial flux is

$$J_{C,r}(r)=\frac{\dot{M}}{4\pi r^2},$$

and conservation across every spherical surface is

$$4\pi r^2 J_{C,r}(r)=\dot{M}.$$

24.2.5 Point-source Green function

For an instantaneous mass M released at the origin in an infinite three-dimensional medium,

$$C(r,t)=\frac{M}{(4\pi D t)^{3/2}}\exp\left(-\frac{r^2}{4Dt}\right), t>0.$$

This solution satisfies

$$\int_{R^3} C(x,t)\, d^3x=M.$$

The characteristic diffusion length and time are

$$\ell_D\sim\sqrt{Dt}, t_D\sim\frac{L^2}{D}.$$

24.3 Proposed geometric-harmonic field G

The field G is intentionally independent of C. Its physical dimensions, observables, and source terms must be declared before a unique field equation can be claimed. A general decomposition may be introduced as

$$G=G_{curv}+G_{pot}+G_{coh}.$$

A possible scalar parametrisation is

$$G_{curv}=\alpha_R R,$$

$$G_{pot}=\alpha_\Phi \Phi_G,$$

$$G_{coh}=\alpha_\Psi |\Psi_G|^2,$$

so that

$$G=\alpha_R R+\alpha_\Phi \Phi_G+\alpha_\Psi |\Psi_G|^2.$$

The coefficients must carry whatever units are required to make all terms compatible.

24.3.1 Candidate field dynamics

A broad nonlinear wave-field ansatz is

$$\frac{\partial^2 G}{\partial t^2} - v_G^2 \nabla^2 G + 2\gamma_G \frac{\partial G}{\partial t} + \kappa_G G + \beta_G G^3 = S_G.$$

Here v_G is a propagation speed, γ_G a damping coefficient, κ_G a linear restoring coefficient, β_G a nonlinear self-interaction coefficient, and S_G a source. Equation (24.23) is a proposed model, not an established law.

A static screened-Poisson limit is

$$-\nabla^2 G + \mu_G^2 G = \rho_G,$$

with spherical exterior solution

$$G(r) = G_\infty + \frac{B}{r} e^{-\mu_G r}.$$

24.4 Coupling C to G

The simplest physically interpretable coupling is to let G alter the effective diffusivity and induce drift. Define

$$D_G = D_G(G, \nabla G, \ldots),$$

and

$$v_G = -\mu_C \nabla U_G,$$

where U_G is an effective potential and μ_C is a mobility. The coupled flux is

$$J_{CG} = -D_G \nabla C + C v_G.$$

The corresponding transport equation is

$$\frac{\partial C}{\partial t} = \nabla \cdot (D_G \nabla C) - \nabla \cdot (C v_G) + S_C.$$

For spherical symmetry,

$$\frac{\partial C}{\partial t} = \frac{1}{r^2} \frac{\partial}{\partial r}\left(r^2 D_G \frac{\partial C}{\partial r} \right) - \frac{1}{r^2} \frac{\partial}{\partial r}\left(r^2 C v_{G,r} \right) + S_C.$$

A convenient test model is

$$D_G(G) = \frac{D_0}{1 + \chi_G G},$$

subject to $1+\chi_G G>0$.

24.5 Definition of C & G harmonic equilibrium

The C & G harmonic equilibrium is defined by vanishing net coupled flux:

$$J_{CG}=0.$$

Hence

$$-D_G\nabla C+C v_G=0,$$

or

$$\nabla \ln C=\frac{v_G}{D_G}.$$

If $v_G=-\mu_C\nabla U_G$ and $D_G=\mu_C k_B T$, then

$$\nabla \ln C=-\frac{\nabla U_G}{k_B T},$$

which integrates to

$$C_{eq}(x)=C_0\exp\left[-\frac{U_G(x)}{k_B T}\right).$$

This is the clearest mathematical expression of the proposed balance: diffusion spreads matter while the effective G-potential organises its equilibrium distribution.

24.6 Harmonic coherence

A phase-coherence measure for two G modes may be introduced as

$$\Gamma_G=\cos(\Delta\phi_G),-1\leq\Gamma_G\leq 1.$$

Perfect phase coherence corresponds to

$$\Gamma_G=1\Leftrightarrow\Delta\phi_G=2\pi n, n\in Z.$$

If the musical perfect-fifth ratio is retained as an analogy or imposed spectral relation, it should be stated separately as

$$\frac{f_G}{f_C}=\frac{3}{2}.$$

Equation (24.39) is not implied by diffusion theory; it is an additional harmonic condition requiring an operational definition of the frequencies f_C and f_G.

24.7 Helispectral combination

A general Helispectral field may be defined as

$$H_S = Z_{NE} Z_{NY} B_{CG}[C, G].$$

A dimensionally valid linear form is

$$H_S = Z_{NE} Z_{NY} \left(a_C C + a_G G + a_{CG} \nabla C \cdot \nabla G \right),$$

where a_C, a_G, and a_{CG} carry the units needed to make the terms commensurate.

A simple product form is

$$H_S = Z_{NE} Z_{NY} \lambda_{CG} C G,$$

where λ_{CG} supplies the required dimensional conversion.

A candidate propagation equation is

$$\frac{\partial^2 H_S}{\partial t^2} - c_H^2 \nabla^2 H_S + 2\gamma_H \frac{\partial H_S}{\partial t} + \frac{d V_H}{d H_S} = S_H.$$

The associated energy-density ansatz is

$$u_H = \frac{1}{2} \left(\frac{\partial H_S}{\partial t} \right)^2 + \frac{c_H^2}{2} \left| \nabla H_S \right)^2 + V_H(H_S).$$

24.8 Dimensionless formulation

Choose reference scales L, $T = L^2 / D_0$, C_0, and G_0, and define

$$\hat{x} = \frac{x}{L}, \tau = \frac{t D_0}{L^2}, \hat{C} = \frac{C}{C_0}, \hat{G} = \frac{G}{G_0}.$$

For constant D_0 and no source,

$$\frac{\partial \hat{C}}{\partial \tau} = \hat{\nabla}^2 \hat{C}.$$

With drift velocity scale U_G the Péclet number is

$$Pe_G = \frac{U_G L}{D_0}.$$

The dimensionless coupled equation becomes

$$\frac{\partial \hat{C}}{\partial \tau} = \hat{\nabla} \cdot \left(\hat{D}_G \hat{\nabla} \hat{C} \right) - Pe_G \hat{\nabla} \cdot \left(\hat{C} \hat{v}_G \right) + \hat{S}_C.$$

24.9 Boundary and initial conditions

A complete problem requires an initial condition

$$C(x, 0) = C_{init}(x),$$

and one or more boundary conditions. Common choices are:

Dirichlet:

$$C \text{¿}_{\partial \Omega} = C_b,$$

Neumann:

$$-D_G \nabla C \cdot n = J_b,$$

Robin:

$$-D_G \nabla C \cdot n = k_m (C - C_b).$$

Equivalent conditions must also be specified for G if Equation (24.23) or (24.24) is used.

24.10 Compact master statement

The mathematical meaning of the C & G Harmonic can be condensed into four equations:

$$\frac{\partial C}{\partial t} + \nabla \cdot J_{CG} = S_C,$$

$$J_{CG} = -D_G \nabla C + C v_G,$$

$$E_G[G] = S_G,$$

$$H_{CG} = Z_{NE} Z_{NY} B_{CG}[C, G].$$

Here E_G denotes the selected and experimentally testable field operator for G.

24.11 Interpretation

The concentration field C answers: **how much dispersed matter is present at each position and time?** The geometric-harmonic field G is proposed to answer: **how is the spatial organisation, effective potential, or harmonic geometry configured?** Their coupling describes how matter disperses through, and may be reorganised by, that geometry.

The rigorous part of the chapter is the conservation-diffusion mathematics for C. The equations for G, the coupling operator B_{CG}, and the factors Z_{NE} and Z_{NY} remain theoretical definitions until their dimensions, measurable observables, parameter values, and falsifiable predictions are established.